

The Stretching Machine: Squares → Parallelograms

Day 11 Enrichment · MathCraft · Grade 6 · Professor Xavier Honablue, M.Ed.

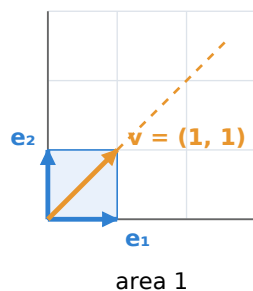
ADVANCED OUTLOOK

Advanced Outlook — not a required 6th-grade standard. This is optional enrichment for curious students. The math you use here is 6th grade (coordinates, area of rectangles, triangles, and parallelograms — 6.G.A.1, 6.G.A.3, 6.NS.C.8). The big idea behind it comes from **linear algebra**, a subject usually taught in college (MIT, Stanford, and Carnegie Mellon all teach it). Nothing on this sheet will be on a test.

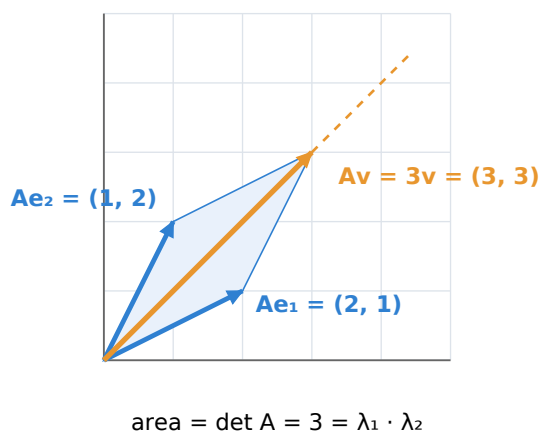
The Big Idea

Imagine a machine called **A** that grabs the whole graph-paper grid and stretches and leans it. Every square on the grid turns into a parallelogram — and every area gets multiplied by the **same number**. Today you will find that number by using only 6th-grade tools.

Before: unit square

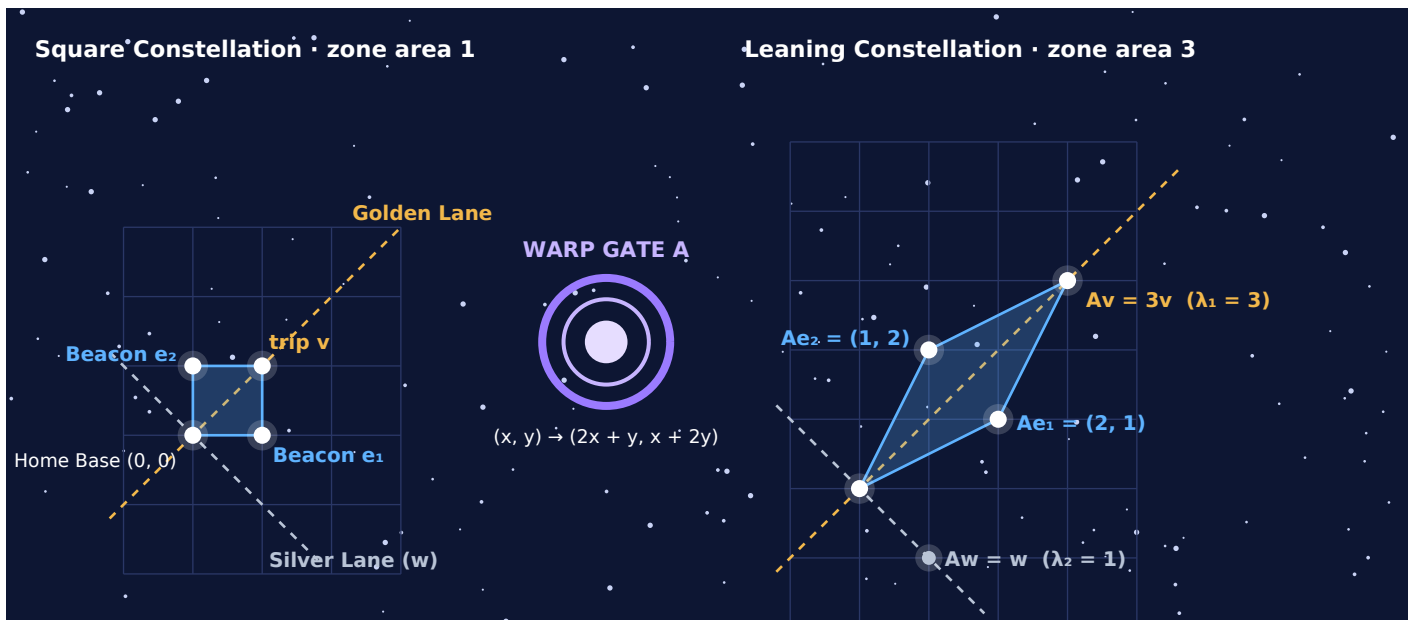


After A: parallelogram



Story Time · The Warp Gate of Sector A

Every symbol in the picture is part of this space story. Read it first, then use the label table as your decoder.



Chapter 1 · The Star Map

Cadet Amara is the navigator of the starship **Horizon**. Her star map is ordinary graph paper. The ship's dock, **Home Base**, sits at the corner $(0, 0)$. Each grid step is one light-year, so every small square of space is one **square light-year**.

Chapter 2 · Two Practice Beacons

Mission Control parks two beacons next to Home Base. **Beacon e_1** ("e-sub-one") is one step **east**, at $(1, 0)$. **Beacon e_2** ("e-sub-two") is one step **north**, at $(0, 1)$. Mathematicians use the letter **e** for these basic one-step arrows, and the little number underneath (the **subscript**) is just the beacon's ID number — beacon #1 and beacon #2.

Chapter 3 · The Square Constellation

Home Base, the two beacons, and one more star at $(1, 1)$ form the **Square Constellation**. The patrol zone inside it has area $1 \times 1 = \mathbf{1 \text{ square light-year}}$.

Chapter 4 · Warp Gate A

Out past the asteroid belt floats **Warp Gate A**. Space that passes through it gets stretched and leaned. The gate follows one rule for every star: $(x, y) \rightarrow (2x + y, x + 2y)$. Amara writes the gate's name in front of a beacon to mean "where it lands after the gate." So **Ae_1** means "Beacon e_1 after Gate A": $(1, 0) \rightarrow (2 \cdot 1 + 0, 1 + 2 \cdot 0) = \mathbf{(2, 1)}$. And **Ae_2** : $(0, 1) \rightarrow \mathbf{(1, 2)}$. The Square Constellation comes out as a **Leaning Constellation** — a parallelogram.

Chapter 5 · The Patrol Zone Grows

Amara uses the box method (Part 2 below) and finds the new patrol zone is **3 square light-years**. She tests a bigger square — it also comes out 3 times bigger. Gate A has an **area power of 3**: every zone that goes through gets 3 times the area. Advanced navigators call this number the **determinant** and write **$\det A = 3$** .

Chapter 6 • The Golden Lane ($\mathbf{Av} = 3\mathbf{v}$)

Captain Brooks makes one trip called $\mathbf{v}$: from Home Base to the star at (1, 1), heading northeast along the **Golden Lane**. Then she flies trip $\mathbf{v}$ through the gate: $(1, 1) \rightarrow (2 + 1, 1 + 2) = (\mathbf{3}, \mathbf{3})$. Look where she lands — **still on the Golden Lane**, just 3 times as far. That is the same as making trip $\mathbf{v}$ three times in a row: $\mathbf{v} + \mathbf{v} + \mathbf{v} = 3\mathbf{v}$. The log entry $\mathbf{Av} = 3\mathbf{v}$ reads: "Gate A turned one trip $\mathbf{v}$ into three trips $\mathbf{v}$." Most routes get **bent** by the gate (Beacon $\mathbf{e}_1$ started due east but came out pointing northeast). The Golden Lane is special: the gate only **stretches** it, never bends it.

Chapter 7 • The Silver Lane

There is one more unbent route: the **Silver Lane**, heading southeast. A trip there called $\mathbf{w}$ goes to (1, -1). Through the gate: $(1, -1) \rightarrow (2 - 1, 1 - 2) = (\mathbf{1}, \mathbf{-1})$. The ship comes out exactly where it would have been. No bend, no stretch: $\mathbf{Aw} = \mathbf{1w}$.

Chapter 8 • Lambda, the Warp Factor (λ)

Navigators give each unbent lane a **warp factor** — how many times longer trips get on that lane. They name it with the Greek letter λ ("lambda," the Greek L — think **L** for **L**ength-changer). The subscripts again are just ID numbers: $\lambda_1 = \mathbf{3}$ for the Golden Lane and $\lambda_2 = \mathbf{1}$ for the Silver Lane.

Chapter 9 • The Captain's Secret

Captain Brooks shares a secret: "Multiply the warp factors of the unbent lanes and you get the gate's area power." $3 \times 1 = 3$. That is the whole bottom line of the picture: **area** = **det A** = $\mathbf{3} = \lambda_1 \cdot \lambda_2$ — the new area, the area power, the number 3, and warp factor times warp factor are all the same. (In college, the unbent lanes are called **eigenvectors** and the warp factors **eigenvalues**.)

Story decoder: star map = coordinate grid · Home Base = (0, 0) · beacons $\mathbf{e}_1, \mathbf{e}_2$ = one-step arrows · Warp Gate A = the machine A · $\mathbf{Ae}_1$ = where $\mathbf{e}_1$ lands · patrol-zone area power = $\det \mathbf{A}$ · Golden Lane = $\mathbf{v}$'s direction · $\mathbf{Av} = 3\mathbf{v}$ = trip $\mathbf{v}$ comes out 3× longer, same lane · Silver Lane = $\mathbf{w}$'s direction · warp factors $\lambda_1 = 3, \lambda_2 = 1$.

What Every Label Means

Read each label out loud the way it is written in the "Say it" column.

Label	Say it	What it means
(x, y)	"x comma y"	An ordered pair : go x units right, then y units up from the starting corner (0, 0).
Before: unit square	"unit square"	The starting shape. Unit means 1, so it is a square with sides of 1 unit, corners at (0, 0), (1, 0), (1, 1), (0, 1).
area 1	"area one"	The unit square covers $1 \times 1 = \mathbf{1}$ square unit .
$\mathbf{e}_1$	"e-sub-one"	The blue arrow pointing right , from (0, 0) to (1, 0). The small ₁ is a subscript : a name tag meaning "first direction." It is NOT an exponent and NOT multiplication.
$\mathbf{e}_2$	"e-sub-two"	The blue arrow pointing up , from (0, 0) to (0, 1). The ₂ means "second direction." $\mathbf{e}_1$ and $\mathbf{e}_2$ are the bottom side and left side of the unit square.
$\mathbf{v} = (\mathbf{1}, \mathbf{1})$	"vee equals one comma one"	The orange arrow . $\mathbf{v}$ is just a name for an arrow we chose to watch. It goes from (0, 0) to (1, 1) — the diagonal of the unit square.
Golden Lane / Silver Lane	—	Story names for the two unbent directions : $\mathbf{v}$'s line (1, 1) and $\mathbf{w}$'s line (1, -1). See Chapters 6–7.

Label	Say it	What it means
orange dashed line	—	The straight path that v points along, drawn longer so you can see its direction . Watch whether v stays on this line after the machine runs.
A (with the gray arrow $\rightarrow$)	"A"	The machine (math name: a transformation or matrix). Its rule: the point (x, y) moves to $(2x + y, x + 2y)$. The arrow means "apply A."
After A: parallelogram	"after A"	What the unit square becomes once A moves every point. Opposite sides are parallel and equal, so it is a parallelogram .
$Ae_1 = (2, 1)$	"A e-sub-one equals two comma one"	Writing A right in front of e_1 means " A applied to e_1 " — where the machine sends the right-arrow. Check: $(1, 0) \rightarrow (2 \cdot 1 + 0, 1 + 2 \cdot 0) = (2, 1)$.
$Ae_2 = (1, 2)$	"A e-sub-two equals one comma two"	Where A sends the up-arrow. Check: $(0, 1) \rightarrow (2 \cdot 0 + 1, 0 + 2 \cdot 1) = (1, 2)$.
light-blue shape & thin blue lines	—	The parallelogram built from Ae_1 and Ae_2 . The thin lines are copies of those two arrows that close up the other two sides. Its 4th corner is $(2, 1) + (1, 2) = (3, 3)$.
$Av = 3v$	"A vee equals three vee"	Where A sends the orange arrow: $(1, 1) \rightarrow (2 + 1, 1 + 2) = (3, 3)$. That is exactly 3 copies of v end to end — stretched 3 times longer but not turned . It stays on the dashed line.
area = det A = 3	"area equals det A equals three"	The parallelogram's area is 3 square units (you'll prove it below). det A (say "det A," short for determinant of A) is the name for the number that tells how many times bigger A makes every area. Advanced Outlook: high school / college.
$\lambda_1 \cdot \lambda_2$	"lambda-sub-one times lambda-sub-two"	λ is the Greek letter lambda . λ_1 and λ_2 are the two stretch amounts along the machine's two special directions: along v 's line everything stretches $3\times$ ($\lambda_1 = 3$); along the line through $(1, -1)$ nothing changes, $1\times$ ($\lambda_2 = 1$). The dot $\cdot$ means multiply: $3 \cdot 1 = 3$. College name: eigenvalues. Advanced Outlook.
the = signs in a chain	—	"area = det A = 3 = $\lambda_1 \cdot \lambda_2$ " says four things are all the same number: the new area, the determinant, 3, and the product of the stretch amounts.

Part 1 · Run the Machine (6th-grade skills: coordinates & expressions)

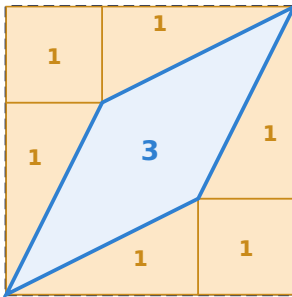
Rule of machine A: $(x, y) \rightarrow (2x + y, x + 2y)$. Fill in the table on graph paper, then plot each new point.

Start point	$2x + y$	$x + 2y$	New point
$e_1 = (1, 0)$	$2(1) + 0 = 2$	$1 + 2(0) = 1$	$(2, 1) \leftarrow$ example
$e_2 = (0, 1)$			
$v = (1, 1)$			
$(2, 0)$			
$(1, -1)$			

Parallelogram Law check: the 4th corner of the new shape should be $Ae_1 + Ae_2$. Add the x's and the y's: $(2 + 1, 1 + 2) = \underline{\hspace{1cm}}$. Is that the same as where v went?

Part 2 · Find the New Area with the Box Method (6.G.A.1)

The parallelogram leans, and its sides are not on grid lines, so base $\times$ height is hard to read here. Instead, **enclose and subtract** (just like Day 7 compound rectangles): draw the smallest grid box around it, then subtract the pieces outside the parallelogram.



Unit square after A

Box: $3 \times 3 = 9$

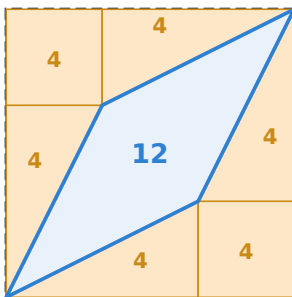
Cut-off pieces: $6 \times 1 = 6$

Parallelogram: $9 - 6 = 3$

- The box is $3 \times 3 = 9$.
- The six orange pieces are two 2-by-1 triangles (area $\frac{1}{2} \times 2 \times 1 = 1$ each), two 1-by-2 triangles (area 1 each), and two 1-by-1 squares (area 1 each): 6 in all.
- Parallelogram area = $9 - 6 = 3$ **square units**. The machine made an area-1 square into an area-3 shape.

Part 3 · Is It Always $\times 3$? Test a Bigger Square

Send the 2×2 square with corners $(0, 0)$, $(2, 0)$, $(2, 2)$, $(0, 2)$ through A. The corners land at $(0, 0)$, $(4, 2)$, $(6, 6)$, $(2, 4)$. Original area = 4. Predict: if A always triples area, the new area should be ____.



2×2 square after A

Box: $6 \times 6 = 36$

Cut-off pieces: $6 \times 4 = 24$

Parallelogram: $36 - 24 = 12$

Box $6 \times 6 = 36$, pieces $6 \times 4 = 24$, so the new area is $36 - 24 = 12 = 3 \times 4$. ✓ **Your turn:** send a 1×2 rectangle — corners $(0, 0)$, $(1, 0)$, $(1, 2)$, $(0, 2)$ — through A, use the box method, and check that the area becomes $3 \times 2 = 6$.

Part 4 · The Special Directions (Advanced Outlook)

Most arrows get **turned** by A (e_1 pointed right, but Ae_1 points up-and-right). Two directions are special:

- **Along $v = (1, 1)$:** $Av = (3, 3) = 3v$. Same direction, 3 times as long. Stretch amount $\lambda_1 = 3$.
- **Along $w = (1, -1)$:** $Aw = (2 - 1, 1 - 2) = (1, -1) = w$. Same direction, same length. Stretch amount $\lambda_2 = 1$.

Multiply the two stretch amounts: $3 \times 1 = 3$ — the same area factor you found with the box method! In college this is written **$\det A = \lambda_1 \cdot \lambda_2$** . You just discovered it with graph paper.

Beyond 6th grade (just for the curious): A is usually written as a grid of numbers called a **matrix**: $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$. Its determinant is $(2 \times 2) - (1 \times 1) = 4 - 1 = 3$. The special directions are called **eigenvectors** and the stretch amounts **eigenvalues**. Engineers use these ideas to design bridges and skyscrapers, and computer programmers use them for video-game graphics and search engines.

Challenge Questions

1. Warp Gate B has the rule $(x, y) \rightarrow (3x, y)$. What does B do to the unit square? What is the new area?
2. Warp Gate C has the rule $(x, y) \rightarrow (x + y, y)$. It slides the top of the square sideways (a **shear**). Plot where the unit square goes. Use base $\times$ height to find its area. Why doesn't the area change? (Hint: Day 8 — cut and slide!)
3. Gate A sends a patrol zone with area 7 square units to a new shape. What is the new area? Explain without drawing.
4. Use the box method to find the area of the parallelogram with corners $(0, 0)$, $(3, 1)$, $(4, 3)$, $(1, 2)$.

Answer Key

Part 1: $e_2 \rightarrow (1, 2)$; $v \rightarrow (3, 3)$; $(2, 0) \rightarrow (4, 2)$; $(1, -1) \rightarrow (1, -1)$. 4th corner = $(3, 3)$ — yes, the same place v went, because $v = e_1 + e_2$.

Part 3 (1 $\times$ 2 rectangle): corners go to $(0, 0)$, $(2, 1)$, $(4, 5)$, $(2, 4)$. Box $4 \times 5 = 20$. Outside pieces: bottom triangle $(0,0),(2,0),(2,1) = 1$; right piece $(2,0),(4,0),(4,5),(2,1) = 6$ (a 2-wide strip with sides 1 and 5: $2 \times (1 + 5) \div 2$); left triangle $(0,0),(0,4),(2,4) = 4$; top piece = $4 \times 1 - 1 = 3$. Total 14. Area = $20 - 14 = 6 = 3 \times 2$. ✓

1. B stretches only sideways: the unit square becomes a 3×1 rectangle, area **3**. **2.** Corners $(0, 0)$, $(1, 0)$, $(2, 1)$, $(1, 1)$: a parallelogram with base 1 and height 1, area **1**. A shear only slides pieces — no area is added or lost. **3.** A multiplies every area by 3: $7 \times 3 = 21$ **square units**. **4.** Box $4 \times 3 = 12$. Outside pieces: bottom triangle $(0,0),(3,0),(3,1) = 1.5$; right piece $(3,0),(4,0),(4,3),(3,1) = 2$; left triangle $(0,0),(0,2),(1,2) = 1$; top piece = $4 \times 1 - 1.5 = 2.5$. Total 7. Area = $12 - 7 = 5$. (Advanced Outlook shortcut: $3 \times 2 - 1 \times 1 = 5$.)

Explore More — Enrichment Websites

MIT, Stanford, and Carnegie Mellon resources are listed first. Items marked **Advanced Outlook** go beyond 6th grade.

MIT	MIT OpenCourseWare 18.06SC — Properties of Determinants (Prof. Gilbert Strang) Advanced Outlook · Free MIT college lecture, notes, and practice on determinants — the "det A" from our picture.
MIT	MIT OpenCourseWare 18.02 — Lecture 3 notes: determinants and area Advanced Outlook · MIT lecture notes showing how a determinant gives the area of a parallelogram.
MIT	Scratch (MIT Media Lab) 6th-grade friendly · Free block coding. Challenge: program a sprite to draw a parallelogram with the pen tool, then a sheared one with the same base and height.
STANFORD	YouCubed at Stanford — Visual Mathematics 6th-grade friendly · Stanford Graduate School of Education resources for seeing math with pictures.
STANFORD	YouCubed at Stanford — Tasks 6th-grade friendly · Free, open-ended math tasks for classes and families.

CARNEGIE MELLON	CMU CS Academy (Carnegie Mellon University) Advanced Outlook · Free graphics-based Python course from Carnegie Mellon. Draw shapes with (x, y) coordinates — try coding the Before/After picture.
OPEN · GR 6	Illustrative Mathematics — Grade 6, Unit 1, Lesson 4: Parallelograms 6th-grade standard · Decompose and rearrange parallelograms to find area.
GEOGEBRA	GeoGebra — Area of a Parallelogram (Discovery) 6th-grade standard · Drag the corners and a slider to watch a parallelogram become a rectangle.
3BLUE1BROWN	3Blue1Brown — The Determinant (lesson page) Advanced Outlook · Animated explanation of how a machine scales area — the exact idea in our picture.